

Lecture 6 (9/25)

Variance, Normal models and Linear Regression

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Related quantities:

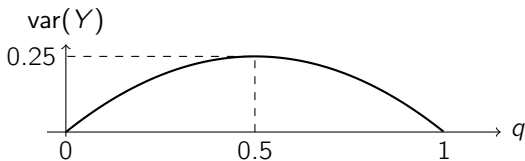
- The standard deviation is $\sigma = \sqrt{\text{var}(Y)}$
- The coefficient of variation is $\text{CV}_Y = \sigma/\mu$ where $\mu = E[Y]$.

Example

Let $Y \sim \text{Bernoulli}(q)$. Compute $\text{var}(Y)$.

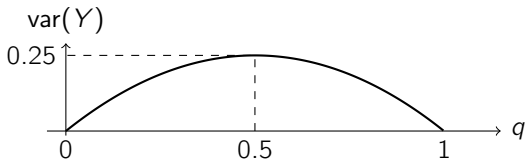
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Let $Y \sim \text{Uniform}(0, L)$. Compute $\text{var}(Y)$ and CV.

The Normal distribution

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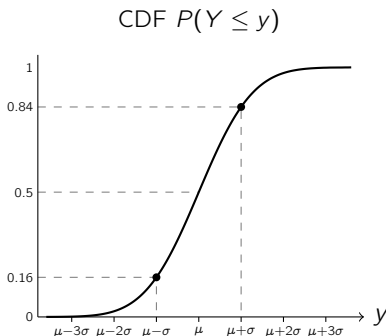
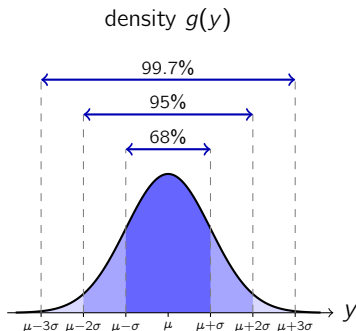
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- If $X \sim \text{Normal}(\mu_X, \sigma_X^2)$ and $Y \sim \text{Normal}(\mu_Y, \sigma_Y^2)$ are independent, then

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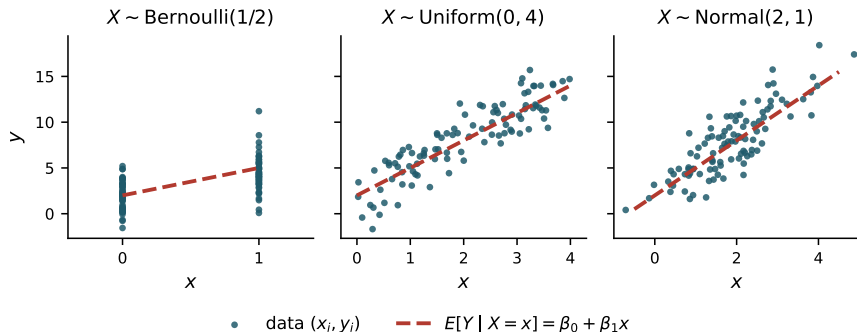
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- X is the predictor
- Y is the response variable
- β_1 is called the regression coefficient of X .
- The variable ϵ may be referred to as an exogenous noise source. Exogeneity means this noise is independent of X .

What data from this model looks like

Data simulated from $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\varepsilon^2)$



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- 3) Homoskedasticity: $\text{var}(Y | X = x)$ does not depend on x .

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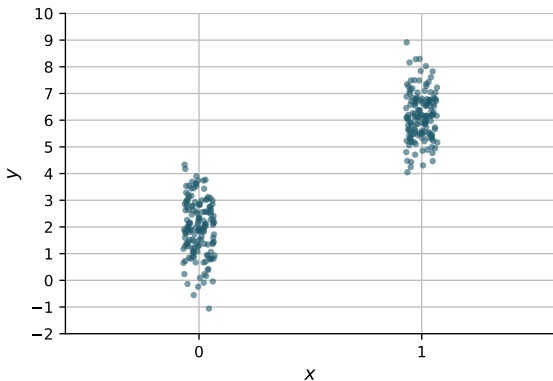
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$$\boxed{\beta_1 = \frac{\text{cov}(X, Y)}{\sigma_X^2}}$$

This formula connects the regression coefficient to the covariance (and hence to expectations we can compute from data).

Example

Write down a linear regression model for the following data, where you can assume $X \sim \text{Bernoulli}(1/2)$. What is $\text{cov}(X, Y)$ and $\text{var}(Y)$?



Example

Two attributes X and Y (e.g. income and height) have a covariance of 2. An increase of X by two units leads to an increase in Y of 1, on average. Assuming a linear regression model $Y | X$, what is the variance of X ?

Marginal variance

Calculate $\sigma_Y^2 = \text{var}(Y)$, where $Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$

$$\text{var}(Y) = \text{var}(\beta_0 + \beta_1 X + \epsilon)$$

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$$\begin{aligned}\text{var}(Y) &= \text{var}(\beta_0 + \beta_1 X + \epsilon) \\ &= \text{var}(\beta_1 X + \epsilon) \quad (\text{adding a constant doesn't change the variance})\end{aligned}$$

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Standardized variables

Let $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$ be two random variables related by a linear regression model.

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Consider the standardized variables

$$Z_X = (X - \mu_X)/\sigma_X, \quad Z_Y = (Y - \mu_Y)/\sigma_Y$$

What is the regression coefficient of Z_X on Z_Y ?

Standardized variables

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Therefore

$$\boxed{\beta_1 \sigma_X / \sigma_Y = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}}$$

This is called the correlation coefficient and denoted ρ .

Solutions

- 1) $Y \sim \text{Bernoulli}(q)$: $\text{var}(Y) = q(1 - q)$.
- 2) $Y \sim \text{Uniform}(0, L)$: $\text{var}(Y) = \frac{L^2}{12}$, $\text{CV} = \frac{L/\sqrt{12}}{L/2} = \frac{1}{\sqrt{3}} \approx 0.58$.
- 3) $\mu_Y = \beta_0 + \beta_1\mu_X$ and $E[XY] = \beta_0\mu_X + \beta_1(\sigma_X^2 + \mu_X^2)$, so $\text{cov}(X, Y) = \beta_1\sigma_X^2$.
- 4) The group means are about ≈ 2 and 6 , so $\beta_1 \approx 4$. $\sigma_X^2 = \frac{1}{4}$ and $\sigma_\epsilon \approx 1$. Hence $\text{cov}(X, Y) \approx \beta_1\sigma_X^2 = 1$ and $\text{var}(Y) \approx \beta_1^2\sigma_X^2 + \sigma_\epsilon^2 = 5$.
- 5) $\beta_1 = \frac{1}{2}$, so $\sigma_X^2 = \text{cov}(X, Y)/\beta_1 = 4$.